Sec: SR.IIT_*COSC(MODEL-A) | GTM-2 | Date: 26-10-24
Time: 3 Hrs | JEE-MAIN | Max. Marks: 300

KEY SHEET

PHYSICS

1	**A**	2	**D**	3	**B**	4	**B**	5	**D**
6	**B**	7	**B**	8	**B**	9	**A**	10	**C**
11	**C**	12	**D**	13	**C**	14	**B**	15	**A**
16	**C**	17	**D**	18	**C**	19	**A**	20	**B**
21	**12**	22	**4**	23	**2**	24	**6**	25	**2**

CHEMISTRY

26	**A**	27	**B**	28	**C**	29	**D**	30	**B**
31	**D**	32	**A**	33	**A**	34	**B**	35	**C**
36	**C**	37	**C**	38	**C**	39	**A**	40	**A**
41	**C**	42	**C**	43	**D**	44	**C**	45	**C**
46	**1**	47	**211**	48	**2**	49	**3**	50	**3**

MATHEMATICS

51	**A**	52	**B**	53	**A**	54	**C**	55	**B**
56	**C**	57	**C**	58	**C**	59	**A**	60	**A**
61	**D**	62	**B**	63	**D**	64	**A**	65	**D**
66	**B**	67	**B**	68	**B**	69	**A**	70	**C**
71	**89**	72	**5**	73	**5**	74	**75**	75	**7**

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SOLUTIONS

PHYSICS

1. As K is variable we take a plate element of Area A and thickness dx at distance x

Capacitance of element

$$dC = \frac{(A)K(1+\alpha x)\varepsilon_0}{dx}$$

Now all such elements are is series so equivalent capacitance

$$\frac{1}{C} = \int \frac{1}{dC} = \int_0^d \frac{dx}{AK\varepsilon_0(1+\alpha x)}$$

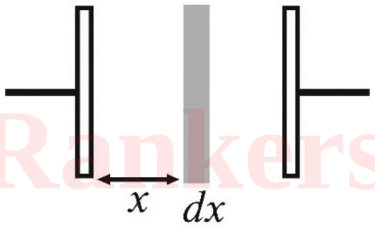
$$\frac{1}{C} = \frac{1}{\alpha AK\varepsilon_0} \ln\left(\frac{1+\alpha d}{1}\right)$$

$$= \frac{1}{C} = \frac{1}{\alpha AK\varepsilon_0}\left(\alpha\, d - \frac{(\alpha d)^2}{2} + \frac{(\alpha d)^3}{3} + \ldots.\right)$$

$$\Rightarrow \frac{1}{C} = \frac{\alpha d}{\alpha AK\varepsilon_0}\left(1 - \frac{\alpha d}{2} + \frac{(\alpha d)^2}{3} + \ldots\right)$$

$$\frac{1}{C} = \frac{d}{AK\varepsilon_0}\left(1 - \frac{\alpha d}{2}\right)$$

$$C = \frac{AK\varepsilon_0}{d}\left(1 + \frac{\alpha d}{2}\right)$$



2. $A = 8\times10^{-3}\ m^2$

$T_1 = 300\ K$

$V_1 = 2.4\times10^{-3}\ m^3$

$V_2 = V_1 + A\Delta x$

$= 2.4\times10^{-3} + 8\times10^{-3}\times0.2$

$= 4\times10^{-3}\, m^3$

$K = 8000\ N/m$

$T_2 = ?$

$P_1 = 10^5\ N/m^2$

$$P_2 = P_0 + \frac{kx}{A} = 10^5 + \frac{8000\times0.2}{8\times10^{-3}} = 3\times10^5\ N/m^2$$

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} = \frac{10^5\times2.4\times10^{-3}}{300} = \frac{3\times10^5\times4\times10^{-3}}{T} = 1500\ K$$

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3. Force on each column $= \frac{mg}{4}$

Strain $= \frac{mg}{4AY}$

$= \frac{50\times 9.8\times 10^5}{4\times\pi(1-0.25)\times 2\times 10^{11}} = 2.6\times 10^{-5}$

4. Net force on particle towards center of circle is

$F_c = \frac{G(4M)^2}{2a^2} + \frac{G(4M)^2}{a^2}\sqrt{2}$

$= \frac{G(4M)^2}{a^2}\left(\frac{1}{2}+\sqrt{2}\right)$

This force will act as centripetal force.

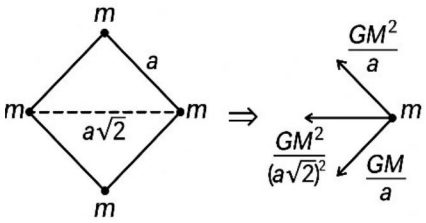
Diatance of particle from center of circle is $\frac{a}{\sqrt{2}}$.

$r = \frac{a}{\sqrt{2}}, F_c = \frac{(4M)v^2}{r}$

$(4M)\frac{v^2}{\frac{a}{\sqrt{2}}} = \frac{G(4M)^2}{a^2}\left(\frac{1}{2}+\sqrt{2}\right)$

$v^2 = \frac{G(4M)}{a}\left(\frac{1}{2\sqrt{2}}+1\right)$

$v^2 = \frac{G(4M)}{a}(1.35); v = 2.32\sqrt{\frac{GM}{a}}$



5. Power factor $= \frac{R}{\sqrt{R^2+X_L^2}} = \frac{R}{\sqrt{R^2+(R)^2}} = \frac{1}{\sqrt{2}}$

Power factor (new) $= \frac{R}{\sqrt{R^2+(X_L-X_C)^2}} = \frac{R}{\sqrt{R^2+(R-R)^2}} = \frac{R}{R} = 1$

$\therefore \frac{\text{New power factor}}{\text{Old power factor}} = \frac{1}{\frac{1}{\sqrt{2}}} = \sqrt{2}$

6. $V_B = \frac{3\times 10^8}{1.5} = 2\times 10^8$

$V_A = 20\times 10^7 + 5\times 10^7 = 2.5\times 10^8$

$\mu = \frac{3}{2,5} = \frac{6}{5} = 1.2$

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7. Given, $|\mathbf{A}\cdot\mathbf{B}|=|\mathbf{A}\times\mathbf{B}|$

Also, we know that and $\mathbf{A}\cdot\mathbf{B}=AB\cos\theta$

$\mathbf{A}\times\mathbf{B}=AB\sin\theta$

From Eqs. (i), (ii) and (iii), we get

$AB\cos\theta=AB\sin\theta$

$\Rightarrow\cos\theta=\sin\theta\Rightarrow\dfrac{\sin\theta}{\cos\theta}=1\Rightarrow\tan\theta=1\Rightarrow\theta=45^\circ$

$\therefore|\mathbf{A}+\mathbf{B}|=\sqrt{A^2+B^2+2AB\cos\theta}$

$=\sqrt{A^2+B^2+2AB\cos\left(45^\circ\right)}=\sqrt{A^2+B^2+\sqrt{2}AB}$

8. Time of flight = 6 + 10 = 16 s

$T=\dfrac{2u\sin\theta}{g}$

$16=\dfrac{2u\sin30^\circ}{10}\Rightarrow u\sin30^\circ=80$

$u\times\dfrac{1}{2}=80\Rightarrow u=160\text{ m/s}$

9. Given, $m=4\text{ kg}$, $l=1\text{ m}$

Breaking tension, $T_{max}=400\text{ N}$

(breaking tension)

Since, stone rotates horizontally,

Thus,

$T_{max}=\dfrac{mv_{max^2}}{l}$

or $v_{max^2}=\dfrac{T_{max}l}{m}$

or $v_{max}=\sqrt{\dfrac{T_{max}}{m}}l=\sqrt{\dfrac{400\times1}{4}}$

or $v_{max}=10\text{ m/s}$

10.

As there is no external torque, angular momentum of system is conserved.

$\Rightarrow I_1\omega_1=I_2\omega_2$

Here I_1 = Moment of inertia of disc of mass M and radius $R=MR^2/2$

$\omega_1=$ Angular speed of disc $1=\omega$

I_2 = Moment of inertia of both discs combined.

$=\dfrac{1}{2}(M+M)\times R^2=MR^2$

So, from Eq. (i)

Or $\dfrac{MR^2}{2}\times\omega=MR^2\times\omega_2$

$\omega_2=\dfrac{\omega}{2}$

11. Acceleration due to gravity on surface of earth,

$g=\dfrac{GM}{R^2}$

At height h acceleration is,

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$$g' = \frac{GM}{(R+h)^2}$$

Dividing Eq. (i) by Eq. (ii), we get

$$\frac{g}{g'} = \frac{(R+h)^2}{R^2} \Rightarrow g' = \frac{gR^2}{(R+h)^2}$$

If mass of body is m, then weight of body at height h is

$$mg' = \frac{(mg)\cdot R^2}{(R+h)^2}$$

Substituting given values, we get weight at height h,

$$w' = mg' = \frac{54\times(6400)^2}{(6400+3200)^2}$$

$$= \frac{54\times6400\times6400}{9600\times9600} = 24$$

12. Given, $A = 24\ m^2$

$\Delta T = 10°C,\ \alpha = 2.0\times10^{-3°}C^{-1}$

We know, $\gamma = 3\alpha = 3\times2\times10^{-3} = 6\times10^{-3°}C^{-1}$

Since, total surface area of cube,

$6a^2 = 24 \Rightarrow a = 2\ m$

$\therefore$ Volume of cube, $V = a^3 = 2^3 = 8\ m^3$

$\Delta V = \gamma V\Delta T = 6\times10^{-3}\times8\times10^6\times10$

$= 4.8\times10^5\ cm^3$

13. Since, $p = aV^{-2}$

$$\frac{dp}{dV} = -2aV^{-3}$$

$$-V\frac{dp}{dV} = \left(-2aV^{-3}\right)(-V)$$

$$-V\frac{dp}{dV} = +2aV^{-2} \Rightarrow -V\frac{dp}{dV} = 2p$$

Bulk modulus $= -V\dfrac{dp}{dV} = 2p$

14. Fundamental frequency of closed organ pipe is,

$f_1 = \dfrac{v}{4L_1}$, v = Speed of sound

L_1 = Length of closed organ pipe

First overtone of open organ pipe has frequency,

$$f_2 = 2\cdot\frac{v}{2L_2}$$

L_2 = Length of open organ pipe

Given, $f_1 = f_2$ and $L_2 = 120$ cm

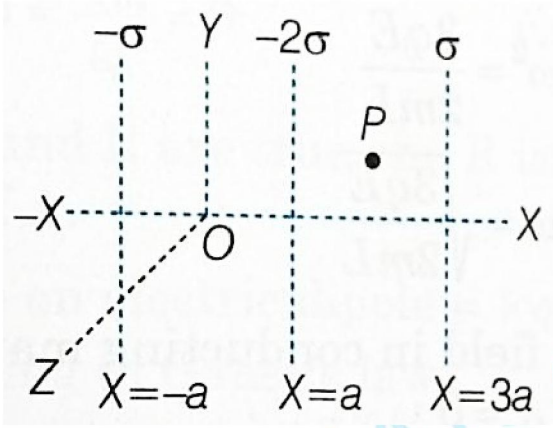
$$\therefore \frac{v}{4L_1} = 2\cdot\frac{v}{2L_2}$$

$$\Rightarrow L_1 = \frac{L_2}{4} = \frac{120}{4} = 30\ cm$$

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15. Electric field at point P

$$E_p = \frac{\sigma}{2\varepsilon_0}(\hat{i}) + \frac{2\sigma}{2\varepsilon_0}(-i) + \frac{3\sigma}{2\varepsilon_0}(-i)$$

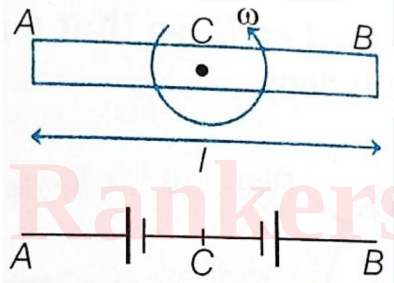


$$= \frac{2\sigma}{\varepsilon_0}(-i)$$

$\therefore x = 2$

16. We know that emf induced across the rotating rod,

$$\varepsilon = \frac{B\omega l^2}{2}$$



$\therefore$ Induced emf in rod AC = induced emf in rod BC

$$= \frac{B\omega(l/2)^2}{2} = \frac{B\omega l^2}{8}$$

Also, by symmetry, $= \frac{B\omega(l/2)^2}{2} = \frac{B\omega l^2}{8}$

$\Rightarrow V_A - V_B = 0$

17. For a resistive load, supply is

$V = 220\sin(50\pi t)$

So, current is

$I = \frac{V}{R}, R = 50\Omega \Rightarrow I = \frac{220}{50}\sin(50\pi t)$

$\Rightarrow I = 4.4\sin(50\pi t)$

$\Rightarrow$ current is at its peak (maximum) when $\sin\theta = 1$

or $50\pi t_1 = \frac{\pi}{2} \Rightarrow t_1 = \frac{1}{100}$ s

Current is half of peak value when $\sin\theta = \frac{1}{2}$

or $50\pi t_2 = \frac{\pi}{6} \Rightarrow t_2 = \frac{1}{300}$ s

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So, time interval, $t_1 - t_2 = \frac{1}{100} - \frac{1}{300}$

$= \frac{2}{300}\,s = \frac{2000}{3000}\,ms = 6.6\,ms$

18. Given, path difference, $\Delta x = \frac{3\lambda}{4}$

If I be the intensity of two identical source, then

$I_{max} = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$

$= I + I + 2I\cos 0^\circ = 4I$

When, $\Delta x = \frac{3\lambda}{4}$

Then, $\Delta\phi' = \frac{2\pi}{\lambda} \times \Delta x = \frac{2\pi}{\lambda} \times \frac{3\lambda}{4} = \frac{3}{2}\pi$

$\therefore$ Intensity $I' = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\Delta\phi'$

$= I + I + 2\sqrt{I \cdot I}\cos\frac{3\pi}{2}$

$= 2I + 2I\cos(\pi + \pi/2)$

$= 2I + 2I(-\cos\pi/2) = 2I$

$\therefore \frac{I'}{I_{max}} = \frac{2I}{4I} = \frac{1}{2}$

19. Relation of stopping potential and threshold wavelength is,

$eV_0 = \frac{hc}{\lambda} - \frac{hc}{\lambda_0}$

Given, when wavelength is λ, $V_0 = 8$ V

$e \cdot 8 = \frac{hc}{\lambda} - \frac{hc}{\lambda_0}$

and when wavelength is 3λ, stopping potential is 2 V,

$e \cdot 2 = \frac{hc}{3\lambda} - \frac{hc}{\lambda_0}$

Dividing Eq. (i) by Eq. (ii), we get

$4 = \frac{\frac{hc}{\lambda} - \frac{hc}{\lambda_0}}{\frac{hc}{3\lambda} - \frac{hc}{\lambda_0}} \Rightarrow \frac{4}{3}\frac{hc}{\lambda} - \frac{4hc}{\lambda_0} = \frac{hc}{\lambda} - \frac{hc}{\lambda_0}$

So, $\lambda_0 = 9\lambda$

20. In a pure inductive circuit current always lags behind the emf by $\frac{\pi}{2}$.

If $\backslash v(t) = v_0 \sin\omega t$ then $\backslash I = I_0 \sin(\omega t - \frac{\pi}{2})$.

Now, given $v(t) = 50\sin(500t)$

And $I_0 = \frac{E_0}{\omega L} = \frac{50}{500 \times 0.02}\left[\because L = 0.02H\right]$

$I_0 = 5\sin\left(500t - \frac{\pi}{2}\right)$

$I_0 = -5\cos(500t)$.

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21. $TV^{\gamma-1} = C$

$\gamma - 1 = n$

$n = \frac{6}{13}$

$\gamma = \frac{4\times\frac{7}{2}R + 2\times\frac{5}{2}R}{4\times\frac{5}{2}R + 2\times\frac{3}{2}R} = \frac{19}{13}$

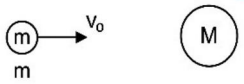
22. (i) $mv_0 = -mv_1 + mv_2$

(ii) $v_0 = v_1 + v_2$

$\frac{2mv_0}{m+M} = v_2$

$KE_f = \frac{1}{2}mv_1^2 = \frac{1}{2}m\left(\frac{M-m}{M+m}\right)^2 v_0^2 = \frac{36}{100}\times\frac{1}{2}mv_0^2$

$\Rightarrow M = 4m$





23. Velocity of particle is given,

$v = 2\sqrt{x}\ m/s$

$\Rightarrow v = \frac{dx}{dt} = 2\sqrt{x} \Rightarrow \frac{dx}{\sqrt{x}} = 2dt$

Integrating we have,

$\int_0^x \frac{dx}{\sqrt{x}} = 2\int_0^t dt$

$2\sqrt{x} = 2t$

So, position of the particle is

$\Rightarrow x = t^2$

$\Rightarrow$ Velocity, $v = \frac{dx}{dt} = 2t$

Acceleration, $a = \frac{dv}{dt} = 2\ m/s^2$

24. x-coordinate of COM,

$x = \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3}$

$= \frac{2m\times0 + 2m\times0 + 2m\times2}{2m+2m+2m} = \frac{4}{6} = \frac{2}{3}$

y-coordinate of COM,

$y = \frac{m_1y_1 + m_2y_2 + m_3y_3}{m_1 + m_2 + m_3} = \frac{2m\times0 + 2m\times2 + 2m\times0}{2m+2m+2m} = \frac{2}{3}$

Position vector of COM is,

$r = x\hat{i} + y\hat{j} = \frac{2}{3}\hat{i} + \frac{2}{3}\hat{j}$

Magnitude of position vector,

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$$|r| = \sqrt{\left(\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^2} = \frac{2\sqrt{2}}{3} = \frac{4\sqrt{2}}{x} \text{ (given)}$$

$\therefore x = 6$

25. Given, thermal resistances

$R_{AB} = R_{CD} = 10.0\ kW^{-1}$

As, C is mid-point of AB, and as we know

$R \propto$ length

$R_{AC} = R_{BC} = 5\ kW^{-1}$

Now, we know that heat current is given as

$Q/t = \Delta T/R$

Let temperature of point C as T.

Consider at point C heat is coming in from A and going out at points B and D.

Therefore, we can write,

$$\frac{T_A - T}{R_{AC}} = \frac{T - T_D}{R_{CD}} + \frac{T - T_B}{R_{BC}}$$

$$\Rightarrow \frac{200 - T}{5} = \frac{T - 125}{10} + \frac{T - 100}{5}$$

$$\Rightarrow 400 - 2T = T - 125 + 2T - 200$$

$$5T = 725°C \Rightarrow T = 145°C$$

The heat current in CD is

$$I_{CD} = \frac{T - T_D}{R_{CD}} = \frac{145 - 125}{10} = \frac{20}{10} = 2\ W$$

Thus, the value of P is 2.

CHEMISTRY

26. Now, during irreversible adiabatic expansion or compression, entropy of system always increase. During adiabatic expansion of ideal gas, if the process is carried out using free expansion, internal energy will remain constant. Should be A.

For an ideal gas: $U = U(T) \Rightarrow \Delta U = nC_V\Delta T$(A):

In adiabatic expansion, q = 0. For free expansion : $q = 0, w = 0 \Rightarrow \Delta U = 0 \Rightarrow \Delta T = 0$.

Temperature does not always decrease

(counterexample : free expansion). $\therefore$ Incorrect.

(B) : Isothermal expansion $\Rightarrow$ T = const (by definition).

(C) : Reversible adiabatic compression $\Rightarrow \Delta S = 0$ (isentropic).

(D) : Irreversible adiabatic compression $\Rightarrow \Delta S = S_{gen} > 0$. Incorrect statement is (A)

27. (A) Highest possible oxidation number: Sc(+3), Ti(+4), V(+5), Cr(+6)

Order: Sc < Ti < V < Cr Correct

(B)Melting points : Ti(1941 K), V(2183 K), Cr(2130 K), Mn(1519 K)

Correct increasing order: Mn < Ti < Cr < V Given order: Ti < V < Cr < Mn $\rightarrow$ Incorrect

(C)Paramagnetic behaviour depends on unpaired electrons : Cu (1), Ni (2), Co (3), Fe (4)

Order : Cu < Ni < Co < Fe Correct

(D)Standard reduction potential $E^\circ\left(M^{2+}/M\right)$:

Mn (–1.18), Fe (–0.44), Co (–0.28)Order: Mn < Fe < Co

Correct Final Answer : Incorrect arrangement is(B)

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28. A) $CH_3CH=CHCH_2-CN$ (i) DIBAL_H (ii) $H_2O \rightarrow CH_3CH=CHCH_2-CHO$

Reason: DIBAL–H reduces nitriles to aldehydes at low T. Correct.

(B) CH_3CH_2-CN (i) $SnCl_2$ / HCl (ii) $H_2O \rightarrow CH_3CH_2-CHO$

Reason: Stephen reduction gives the aldehyde. Correct.

(C)$CH_3CH_2COCl+(CH_3)_2Cd \rightarrow CH_3COCH_3+CdCl_2$

Reason: Diorganyl cadmium adds one CH_3 to the acyl chloride to give a ketoneretaining the ethyl group.

Actual product : $\mathbf{CH_3CH_2COCH_3}$ (2- butanone). Given product acetone is wrong.

Incorrect match.

(D) $C_2H_5COCH_3+NaOCl \longrightarrow C_2H_5COO^-Na^++CHCl_3$

Reason: Haloform reaction of a methyl ketone with hypochlorite. Correct.

Final Answer : (C) is not correctly matched.

29. Given : P10 mL0.1MNaOH + 5 mL0.1MHCl

Q10 mL0.1MNaOH +15 mL0.1MCH3COOH

R10 mL0.1MNH3 +10 mL0.1MNH4Cl

S10 mL0.05MNaF + 5 mL0.1MHF

Buffer criterion :

mixture must contain a weak acid and its conjugate base, or a weak base and its conjugate acid, both present after mixing.

Pn_{NaOH} $0.010L\times0.1M=0.001mol$ n_{HCl} $0.005\ L\times0.1M=0.0005$ mol

Excess strong base 0.0005 mol remains → strong base present Not a buffer.

Qn_{NaOH} $0.001mol$ n_{CH_3COOH} 0.0015 mol

After neutralization acetate 0.001 mol and acetic acid 0.0005 mol remain

Weak acid plus conjugate base → buffer.

RWeak base NH_3 with its salt NH_4Cl in comparable amounts

Directly a buffer.

Sn_{NaF} $0.010L\times0.05M=0.0005$ mol n_{HF} $0.005L\times0.1M=0.0005$ mol

HF and $F^{\ominus}$ present together → buffer.

Final Answer : Q,R,S only (option D)

30. (A) Secondary alkyl fluoride with $NH_2^{\ominus}$ and

heat Elimination of alkyl fluorides often gives the Hofmann(less substituted) alkene because the E2 transition state has carbanion character and *F* withdraws strongly.Not Saytzeff.

(B) 1–Chloropropane with alcoholic KOH and heatSmall strong base, –elimination → propene (only alkene possible, which is the more substituted for any alternative). Saytzeff.

(C) $t-BuO^-K^+$ removeleast hindered *H* gives the Hofmann(less substituted).

(D) Quaternary ammonium salt $(-\overset{\oplus}{N}Me_3)$ with RO^- and heat Hofmann elimination: less substituted alkene dominates.

Not Saytzeff. Final Answer:(B).

31. (A) $CHCl_3+CH_3COCH_3 \rightarrow$ strong H–bonded complex forms → contraction

(B) $H_2O+HCl \rightarrow$ strong solvation → contraction (C)$H_2O+HNO_3 \rightarrow$ strong solvation → contraction

(D) $H_2O+C_2H_5OH \rightarrow$ positive deviation from Raoult's law → no contraction

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32. (A) $(CH_3)_3CH-Cl_2, h\nu \rightarrow$ primary chloride major due to selectivity and reactivity principal $\rightarrow$ incorrect

(B) $CH_2=CH-CH_2-CH_3-NBS, \Delta \rightarrow$ allylic bromide via radical allylic substitution $\rightarrow$ correct

(C) $CH_2=CH-CH_3-(i)BH_3\cdot THF$ (ii) $H_2O_2, OH^{\ominus} \rightarrow CH_3-CH_2-CH_2OH$ (anti – Markovnikov) $\rightarrow$ correct

(D) $CH_3-CH=CH_2-HCl$, peroxide $\rightarrow$ Peroxideeffect not applicable in HCl Markovnikov addition;

this effect occurs with HBr only, not with HCl $\rightarrow$ correct

33. Electrolysis of $CuSO_4$

At anode : $2H_2O \longrightarrow O_2 + 4H^{\oplus} + 4e^{\ominus}$

$n(O_2) = 5.6\ L \div 22.4\ L\ mol^{-1} = 0.25\ mol$

Charge passed = 4F per mol $O_2 = 4 \times 0.25 = 1\ F$

At cathode: $Cu^{2\oplus} + 2e^{\ominus} \rightarrow Cu$

Moles $Cu = (1\ F) \div (2\ F\ per\ mol) = 0.5\ mol$

Mass $Cu = 0.5 \times 63.5 =$ **31.75 gg**

34. Acidic strength order

Aniline $\rightarrow (NaNO_2 / HCl, 0-5°C) \rightarrow$ diazonium salt P

$P \rightarrow (H_2O, warm) \rightarrow$ phenol Q

$Q \rightarrow$ (dil. $HNO_3) \rightarrow$ **R = o**–nitrophenol (intramolecular H–bond) + **S = p**–nitrophenol (intermolecular H–bond)

$R \rightarrow$ (conc. $HNO_3) \rightarrow$ T(picric acid, 2,4,6–trinitrophenol)

Acidity: T (strongly stabilized phenoxide) > S (para, no intramolecular H–bond, better solvation) >R (ortho, intramolecular H - bond stabilizes neutral form) > Q (phenol)

Order: T > S > R > Q (option A)

35. Entropy change for vaporization of 3.6 g water at 100°C

Given: Molar heat of vaporization, $\Delta H_{vap} = 40.85\ kJ\ mol^{-1}$

Mass = 3.6

Molar mass of $H_2O = 18\ g\ mol^{-1}$ $T = 373\ K$

Number of moles: $n = 3.6/18 = 0.2\ mol$

Entropy change for vaporization :

$\Delta S = (n \times \Delta H_{vap}) / T\Delta S = (0.2 \times 40.85 \times 10^3\ J\ mol^{-1}) / 373\ K\Delta S$

$= 8.17 \times 10^3 / 373\Delta S = 21.89 J\ K^{-1}$

36. (B) NF_3 (0.23) < HI (0.38) < H_2S (0.95) < $CHCl_3$ (1.04)

37. If Hund's rule is violated, which becomes diamagnetic from paramagnetic?

$C_2 \rightarrow 12e^-$ (diamagnetic already)

$CN^- \rightarrow 14e^-$ (diamagnetic)

$N2^{2-} \rightarrow 16e^-$ like

O_2; normally two unpaired electrons in π^* orbitals (paramagnetic).

If Hund's rule is violated, both pair in one π^* orbital $\rightarrow$ diamagnetic.

$BN \rightarrow 12e^-$ (diamagnetic)Answer: $N2^{2-}$ (option C)

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38. Cell: $Ag\,|\,AgBr(s)\,|\,Br^-(0.1M)\,||\,Ag+(0.1M)\,|\,Ag$,

$E_{cell} = 0.643\ V, \dfrac{2.303RT}{F} = 0.06$

Left half–cell is $Ag\,|\,Ag^+$ buffered by $AgBr(s) : AgBr(s) \rightleftharpoons Ag^+ + Br^-$

$K_{sp} = \left[Ag^+\right]\left[Br^-\right] \Rightarrow \left[Ag^+\right]_L = \dfrac{K_{sp}}{\left[Br^-\right]} = \dfrac{K_{sp}}{0.1}$

Right half–cell: $\left[Ag^+\right]_R = 0.1M$

Concentration cell: $E_{cell} = 0.06\log\left(\dfrac{\left[Ag^+\right]_R}{\left[Ag^+\right]_L}\right)$

$= 0.06\log\left(\dfrac{0.1}{K_{sp}/0.1}\right) = 0.06\log\left(\dfrac{0.01}{K_{sp}}\right)$

$\Rightarrow \dfrac{E_{cell}}{0.06} = \log(0.01) - \log K_{sp} = -2 - \log K_{sp}$

$\log K_{sp} = -\dfrac{E_{cell}}{0.06} - 2[6pt]$ With $E_{cell} = 0.643\ V$:

$\log K_{sp} = -\dfrac{0.643}{0.06} - 2 = -12.72$

39. For fixed permanent dipoles: $U_{dd} \propto \dfrac{1}{r^3}$.

(A) both have nonzero dipole ($\mu \neq 0$) $\Rightarrow$ dipole–dipole ($1/r^3$).
(B) Polar and non Polar
(C)Both are non polar $\Rightarrow$ dispersion ($1/r^6$).
(D) electrolytes: ion–ion (1/r).
1) Polar and polar (permanent dipoles)

Fixed orientation: $U_{dd} = \dfrac{1}{4\pi\varepsilon_0 r^3}\left[\mu_1\cdot\mu_2 - 3\left(\mu_1\cdot\hat{r}\right)\left(\mu_2\cdot\hat{r}\right)\right] \propto \dfrac{1}{r^3}$

Thermally averaged (Keesom): $\langle U\rangle = -\dfrac{\mu_1^2\mu_2^2}{3\left(4\pi\varepsilon_0\right)^2 k_B T r^6} \propto \dfrac{1}{Tr^6}[6pt]$

2) Polar and nonpolar (dipole induced dipole; Debye) $\langle U\rangle \propto -\dfrac{\mu^2\alpha}{\left(4\pi\varepsilon_0\right)^2 r^6} \propto \dfrac{1}{r^6}$

($\mu =$ permanent dipole, $\alpha =$ polarizability)
3) Nonpolar n nonpolar (Londondispersion)

$\langle U\rangle = -\dfrac{C_6}{r^6}, C_6 \propto \dfrac{\alpha_1\alpha_2 I_1 I_2}{I_1 + I_2} \propto \alpha_1\alpha_2$, so $\langle U\rangle \propto \dfrac{1}{r^6}$

40. (i) $\left[Zn(gly)_2\right]$: Zn(II) is 4–coordinate (tetrahedral); two unsym bidentate N, O ligands

$\Rightarrow \left[M(AB)_2\right]$ is chiral (no mirror plane) $\Rightarrow$ optically active (?)

(ii) [PdClBr(gly)]– : Pd(II) is square–planar d^8; bidentate gly occupies cis sites. Cl and Br are trans, but two arrangements exist (Cl trans to N or to O) $\Rightarrow$ geometrical isomerism (?)

(iii) $\left[IrBr_4(en)\right]^-$: octahedral with one en + four identical Br ; only one arrangement $\Rightarrow$ no cis – trans (?)

(iv) $\left[RhCl_2\left(NH_3\right)_4\right]^+$: octahedral $MA_4B_2 \Rightarrow$ cis, trans (2 GI); no optical isomer.

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So stereoisomers = geometrical isomers = 2

41. Reductive ozonolysis:

$RCH=CH_2 \xrightarrow[Zn/H_2O]{O_3} RCHO + HCHO$

Choose X so both products lack $\alpha - H$ (Cannizzaro positive).

$X=(CH_3)_3C-CH=CH_2 \Rightarrow$

$Y=(CH_3)_3C-CHO(\text{no } \alpha-H),\ Z=HCHO\ (\text{no}\alpha-H)$

Other options give at least one aldehyde with $\alpha - H$ (no Cannizzaro).

$X=(CH_3)_3C-CH=CH_2$

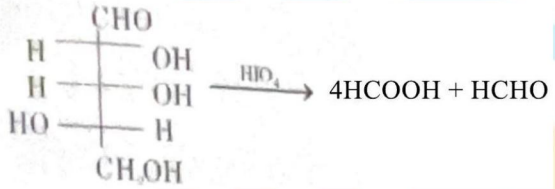
42. (A) $F^{\ominus}$ vs $Cl^{\ominus}$: $EA(Cl) > EA(F) \Rightarrow$ extra $e^{\ominus}$ is held tighter in $Cl^{\ominus}$, so $IE(F^{\ominus}) < IE(Cl^{\ominus})$. (Statement false)

(B)Group –13 anomaly: due to poor d/f shielding & inert – pair effect, IE(Tl) > IE(Ga). (Statement false)

(C)Group–14: IE rises slightly from Sn to Pb (inert–pair effect) ⇒ Sn < Pb. (True)

(D)For anions, $EA(S) > EA(O) \Rightarrow$ removing $e^{\ominus}$ from $S^{\ominus}$ is harder, so$IE(O^{\ominus}) < IE(S^{\ominus})$. (Statement false)

43.



44. $CFSE_o = -1.2\Delta_o$ and increases with ligandfield/chelation(en > NH_3).

$[Ni(NH_3)_6]^{2+} < [Ni(en)_2(NH_3)_2]^{2+} < [Ni(en)_3]^{2+}$.

(A) Wrong: square planar $[PtCl_4]^{2-}, [PdCl_4]^{2-} \gg$ tetra $[NiCl_4]^{2-}$.

(B) Wrong: $[Fe(H_2O)_6]^{3+} > [FeF_6]^{3-}$.

(D) Wrong: square planar $[Ni(CN)_4]^{2-} \gg$ octa $[Ni(NH_3)_6]^{2+}$.

45. Use dG = –SdT + VdP

$\Rightarrow \left(\frac{\partial G}{\partial T}\right)_P = -S, \left(\frac{\partial G}{\partial P}\right)_T = V$

$C_P = \left(\frac{\partial H}{\partial T}\right)_P,\ C_V = \left(\frac{\partial U}{\partial T}\right)_V$

46. Number of incorrect statements = 1
(only C)

(A) True: Zero order $t_{1/2} = \frac{[A]_0}{2k} \Rightarrow$ as $[A]_0 \downarrow, t_{1/2} \downarrow$.

(B) True: A reactant can have order 0 ⇒ rate independent of its concentration.

(C) False: Order can be fractional; molecularity (elementary step) is an integer.

(D) True: k_0 (zero order) $= molL^{-1}s^{-1}$, k_2 (second order) $= Lmol^{-1}s^{-1}$

47. From (i): $\Delta H_1 = -13.7 = [\Delta_f H^{\circ}(KCl(aq)) + (-68.4)] - [(-116.5) + (-39.3)]$

$\Rightarrow \Delta_f H^{\circ}(KCl(aq)) = -101.1 kcalmol^{-1}$

From (v):

$4.4 = \Delta_f H^{\circ}(KCl(aq)) - \Delta_f H^{\circ}(KCl(s)) \Rightarrow \Delta_f H^{\circ}(KCl(s)) = -101.1 - 4.4 = -105.5 kcalmol^{-1}$

$|10X| = 211$

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48. Colourless aq. solutions:
ions with no d– –d/CT absorption.

$CuSO_4 : Cu^{2+}\left(d^9\right)$ blue ;

$ZnSO_4 : Zn^{2+}\left(d^{10}\right)$ colourless ;

$FeSO_4 : Fe^{2+}\left(d^6\right)$ pale green ;

$Cr_2\left(SO_4\right)_3 : Cr^{3+}\left(d^3\right)$ coloured ;

NaCl : colourless ;

$KMnO_4$: purple ;

$K_2Cr_2O_7$: orange .

Number colourless $= 2\left(ZnSO_4, NaCl\right)$.

49. (ii)

(ii) 

50. Let $r = k[A]^x[B]^y[C]^z$.

Doubling A doubles rate: $2^x = 2 \Rightarrow x = 1$.

Doubling B has no effect: $2^y = 1 \Rightarrow y = 0$.

Tripling C makes rate $\times 9 : 3^z = 9 = 3^2 \Rightarrow z = 2$

Overall order = x + y + z = 1 + 0 + 2 = 3.

MATHEMATICS

51. $\dfrac{dy}{dx} = \dfrac{(x-2)+(y-2)}{(x-2)-(y-2)}$

put $x - 2 = h$, $y - 2 = k$

$\dfrac{dk}{dh} = \dfrac{h+k}{h-k}$ put $k = vh$

$\Rightarrow v + h\dfrac{dv}{dh} = \dfrac{1+v}{1-v}$

$\Rightarrow \int \dfrac{1-v}{1+v^2}dv = \int \dfrac{dh}{h} \Rightarrow \tan^{-1} v = \dfrac{1}{2}\ln\left(1+v^2\right) + \ln h + c$

$\Rightarrow \tan^{-1}\left(\dfrac{y-2}{x-2}\right) = \dfrac{1}{2}\ln\left(1+\dfrac{(y-2)^2}{(x-2)^2}\right) + \ln(x-2) + c$

$\because$ it passes through (3, 2)

$\Rightarrow \tan^{-1} 0 = \dfrac{1}{2}\ln 1 + \ln 1 + c \Rightarrow c = 0$

$\because$ it also passes through (p + 2, 3)

$\tan^{-1}\left(\dfrac{1}{p}\right) = \dfrac{1}{2}\ln\left(1+\dfrac{1}{p^2}\right) + \ln p$

$2\tan^{-1}\left(\dfrac{1}{p}\right) = \ln\left(1+p^2\right)$

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52. $T_n = \frac{((n+1)x+3)-(nx+3)}{((n+1)x+3)(nx+3)} = \frac{1}{nx+3} - \frac{1}{(n+1)x+3}$

$S_{12} = V_1 - V_2 + V_2 - V_3 + \ldots\ldots V_{12} - V_{13} = V_1 - V_{13}$

$= \frac{1}{x+3} - \frac{1}{13x+3}$

$\lim_{x\to 1} \frac{1}{4} - \frac{1}{16} = \frac{3}{16}$

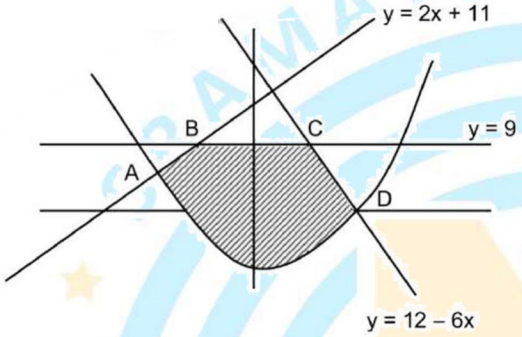
53. $x^2 - 4 = 2x + 11$

$x = -3, 5 \therefore A(-3, 5)$

$2x + 11 = 9$

$x = 1 \therefore B(-1, 9)$

$C\left(\frac{1}{2}, 9\right)$ and $D(2, 0)$



Required area $\int_{-3}^{-1} (2x+11) - (x^2-4)\,dx + \int_{-1}^{1/2} 9 - (x^2-4)\,dx + \int_{1/2}^{2} 12 - 6x - (x^2-4)\,dx$

$= \int_{-3}^{-1} (2x - x^2 + 15)\,dx + \int_{-1}^{1/2} (13 - x^2)\,dx + \int_{1/2}^{2} (16 - 6x - x^2)\,dx = \frac{511}{12}$

54. $\int_0^1 \left(f^4(x) - 8x^2 \cdot f^2(x) + 16x^4\right)dx \Rightarrow \int_0^1 \left(f^2(x) - 4x^2\right)^2 = 0 \Rightarrow f^2(x) = 4x^2$

55. All the coefficient of odd powered terms, is an odd number of odd degree terms multiplied together and all coefficient of even powered terms has an even number of odd degree terms multiplied together. Since every odd degree term is negative and every even degree is positive so we sum is just $Q(-1) = P(-1)^5 = \frac{243}{32}$.

56. Assume AB is on the line y = 2x – 17 and coordinate of other two are on parabola are $C(x_1, y_1)$ and $D(x_2, y_2)$, then CD is on the line L whose equation is y - 2x = b by solving with curve we get

$x^2 - 2x - b = 0 \Rightarrow x_1, x_2 = 1 \pm \sqrt{b+1}$

assume length of one side of square is a

$\Rightarrow a^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$

$= 5(x_1 - x_2)^2 = 20(b+1)$

Now $a = \left|\frac{17+b}{\sqrt{5}}\right|$

$\Rightarrow b = 3$ or $63 \Rightarrow a^2 = 80$ or 1280.

57. Let $\sqrt[3]{x^2+2x} = y = f(x) \quad (f(2) = 2)$

$x = -1 + \left(y^3 + 1\right)^{1/2}$

$I = \int_0^2 \left(f^{-1}(x) + f(x) + 1\right)dx$

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consider $\int_0^2 f^{-1}(x)dx = \int_0^2 t\ f'(t)dt$; Let $f^{-1}(x) = t; x = f(t); dx = f'(t)dt$

$= [t \cdot f(t)]_0^2 - \int_0^2 f(t)dt = 2f(2) - \int_0^2 f(x)dx$

Now; $I = \int_0^2 \left(f(x) + f^{-1}(x)\right)dx + 2 \Rightarrow I = 2f(2) + 2 = 6$

58. $I = \int \frac{x^{14} + 2x^{11}dx}{\left(x^6 + x^3 + 1\right)^3} + \int \frac{x^5 - x^{11}dx}{\left(x^6 + x^3 + 1\right)^3}$

$I = \int \frac{x^{-4} + 2x^{-7}}{\left(1 + x^{-3} + x^{-6}\right)^3}dx + \int \frac{x^{-4} - x^2}{\left(x^3 + x^{-3} + 1\right)^3}dx$

$1 + x^{-3} + x^{-6} = t$ $\qquad$ $x^3 + x^{-3} + 1 = u$

$\left(-3x^{-4} + -6x^{-7}\right)dx = dt$ $\qquad$ $\left(3x^2 - 3x^{-4}\right)dx = du$

$I = -\frac{1}{3}\int \frac{dt}{t^3} - \frac{1}{3}\int \frac{du}{u^3}$

$I = \frac{1}{6\left(1 + x^{-3} + x^6\right)^3} + \frac{1}{6}\frac{1}{\left(x^3 + x^{-3} + 1\right)^3} + c$

59. $\tan^{-1}\left(\frac{y-2}{x-7}\right) = \tan^{-1}\frac{1}{2} \Rightarrow y - 2 = \frac{1}{2}(x-7)$

$\Rightarrow x - 2y - 3 = 0 \quad --(1)$

where $x > 7$ and $y > 2$

Equation (2) is a circle with diametric ends (–2, 1) and (0, 5).

Equation of circle: $x^2 + y^2 + 2x - 6y + 5 = 0$

Here, length of $\perp$ from C is equal to radius, but $x < 7$

So, no solution.

60. $\frac{\sum_{i=1}^{n} x_i f_i}{\sum_{i=1}^{n} f_i} = \frac{15a + 25b + 980}{26 + a + b} = \frac{277}{9} \qquad \ldots(i)$

Also $30 + \frac{\frac{26 + a + b}{2} - (a + b + 3)}{11} \times 10 = \frac{335}{11}$

$\Rightarrow a + b = 19 \qquad \ldots(ii)$

From (1) and (2) $a = 7, b = 12$.

61. $\frac{1}{b_n + 1} + \frac{1}{b_{n+1} - 1} = \frac{1}{b_n - 1}$ using this relation

$\Rightarrow \frac{1}{b_0 + 1} + \frac{1}{b_1 - 1} = \frac{1}{3} + \frac{1}{4} = \frac{7}{12}$

62. $y = \frac{1 + \cos x + \cos^2 x}{1 + \cos x} = \frac{(1 + \cos x)^2 - \cos x}{1 + \cos x}$

$y = (1 + \cos x) - \left(\frac{\cos x + 1 - 1}{1 + \cos x}\right) = (1 + \cos x) - 1 + \frac{1}{1 + \cos x}$

Non $(1 + \cos x) + \frac{1}{1 + \cos x} \geq 2$

$\therefore (1 + \cos x) + \frac{1}{(1 + \cos x)} - 1 \geq 1$

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63. $f(x)=(a-2)x^3+3x^2-bx$ is symmetric about line x = 3, $\Rightarrow$ it has to be quadratic $\Rightarrow$ a = 2

hence quadratic is $f(x)=3x^2-bx$

So $-\frac{b}{2a}=3\Rightarrow\frac{b}{2\times3}=3\Rightarrow b=18$

Min value of $f(x), c=-\frac{D}{4a}$

$c=-27$

$\therefore a+3b+2c=2+54-54=2$

64. $\bar{a},\bar{b},\bar{c},\bar{d}$ be co-planar

$\Rightarrow \bar{a}-\bar{b},\bar{a}-\bar{c},\bar{a}-\bar{d}$ are co-planar

$\therefore(\bar{a}-\bar{b})\cdot\{(\bar{a}-\bar{c})\times(\bar{a}-\bar{d})\}=0$

$(\bar{a}-\bar{b})\cdot\{\bar{d}\times\bar{a}+\bar{a}\times\bar{c}+\bar{c}\times\bar{d}\}=0$

$\therefore\left[\bar{a}\ \bar{c}\ \bar{d}\right]-\left[\bar{b}\ \bar{d}\ \bar{a}\right]-\left[\bar{b}\ \bar{a}\ \bar{c}\right]-\left[\bar{b}\ \bar{c}\ \bar{d}\right]=0$

$\left[\bar{a}\ \bar{c}\ \bar{d}\right]+\left[\bar{a}\ \bar{d}\ \bar{b}\right]+\left[\bar{a}\ \bar{b}\ \bar{c}\right]=\left[\bar{b}\ \bar{c}\ \bar{d}\right]$

65. Total face of cubes which have coloured face (s) = 120

No of faces have 3 colour = 8

$\therefore$ Prob of 2 more faces colored given that one is coloured already given by $=\frac{8}{120}=\frac{1}{15}$

66. No of ways

$=$ coefficient of α^8 in $\left(1+\alpha+\alpha^2+\alpha^3+\alpha^4+\alpha^5+\alpha^6-\alpha^4\right)^6$

$=$ Coefficient of α^8 in $\left(\frac{1-\alpha^7}{1-\alpha}-\alpha^4\right)^6$

$=$ Coefficient of α^8 in $\left(\frac{1-\alpha^7}{1-\alpha}\right)^6-{}^6C_1\cdot\left(\frac{1-\alpha^7}{1-\alpha}\right)^5\alpha^4+{}^6C_2\left(\frac{1-\alpha^7}{1-\alpha}\right)^4(\alpha^8)+\ldots.$

$=$ coefficient of α^8 in $\left(1-\alpha^7\right)^6(1-\alpha)^{-6}-{}^6C_1\cdot\left(1-\alpha^7\right)^5(1-\alpha)^{-5}\cdot\alpha^4+{}^6C_2\left(1-\alpha^7\right)^4(1-\alpha)^{-4}\alpha^8+\ldots.$

$={}^{13}C_8-{}^6C_1\cdot{}^6C_1-{}^6C_1{}^8C_4+{}^6C_2$

67. $T_r=\frac{1}{\sin\left(2^r\right)^\circ}$

$=\operatorname{cosec}\left(2^r\right)^\circ=\cot\left(2^{r-1}\right)^\circ-\cot\left(2^r\right)^\circ$

$S=\sum_{r=1}^{7}T_r=\cot1^\circ-\cot(128)^\circ$

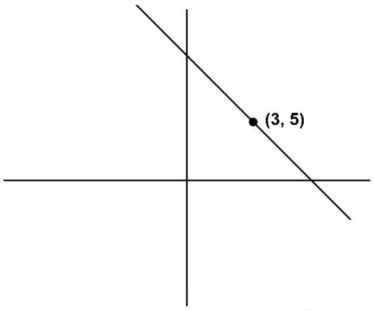
$=\frac{\cos1^\circ}{\sin1^\circ}-\frac{\cos128^\circ}{\sin128^\circ}$

$=\frac{\sin\left(127^\circ\right)}{\sin1^\circ\sin128^\circ}$

$=\frac{\sin53^\circ}{\sin1^\circ\sin52^\circ}$

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68. with the co-ordinate axes.



min area in 1st quad will occur when (3, 5) is a mid point.

$=\frac{1}{2}\times 6\times 10$

$=30$

So Δ will form in 2nd & 4th quadrant only

69. C_1 : Centre $(2,4)$, $r_1 = r$ $(r > 0)$

C_2 : Centre $(5,1)$, $r_2 = 4$

$C_1C_2 = \sqrt{(5-2)^2+(1-4)^2} = 3\sqrt{2}$

$|r_1 - r_2| < C_1C_2 < r_1 + r_2$

$|r-4| < 3\sqrt{2} < r+4$

$|r-4| < 3\sqrt{2}$ and $3\sqrt{2} < r+4$

$-3\sqrt{2} < r-4 < 3\sqrt{2}$

$4-3\sqrt{2} < r < 4+3\sqrt{2}$

$\Rightarrow 0 < r < 4+3\sqrt{2}$ and $r > 3\sqrt{2}-4$

$\Rightarrow 3\sqrt{2}-4 < r < 4+3\sqrt{2}$

70. $a\left(-\frac{3x+1}{x-1}\right)^2 + b\left(-\frac{3x+1}{x-1}\right) + c = 0$

$\alpha = -\frac{3x+1}{x-1}$

$\alpha x - \alpha = -3x - 1$

$x = \frac{\alpha-1}{\alpha+3} \Rightarrow$ roots are $\frac{\alpha-1}{\alpha+3}, \frac{\beta-1}{\beta+3}$

71. 89

$S_n = \frac{1}{\sin 1^\circ}\sum_{k=22}^{66}\frac{\sin((2k+2)^\circ-(2k+1)^\circ)}{\sin(2k+1)^\circ\sin(2k+2)^\circ}$

$S = \frac{1}{\sin 1^\circ}\sum_{k=22}^{66}\left(\cot(2k+1)^\circ - \cot(2k+2)^\circ\right)$

$S = \frac{1}{\sin 1^\circ}\left(\cot 45^\circ - \cot 46^\circ + \cot 47^\circ - \cot 48^\circ \ldots + \cot 133^\circ - \cot 134^\circ\right)$

Here $\cot 47^\circ - \cot 133^\circ = 0$

$S = \frac{1}{\sin 1^\circ}\cdot\left(\cot 45^\circ - \cot 90^\circ\right) = \sec 89^\circ$

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72. $I=\int_1^2 \frac{1}{x^3\left(x^2+2\right)^2}\,dx=\int_1^2 \frac{1}{x^7\left(1+\frac{2}{x^2}\right)^2}\,dx=-\frac{1}{4}\int_3^{3/2}\left(\frac{t-1}{2}\right)^2\frac{dt}{t^2}$ where $t=1+\frac{2}{x^2}$

$I=\frac{1}{4}\int_{3/2}^3 \frac{t^2+1-2t}{4}\frac{1}{t^2}dt=\frac{1}{16}\int_{3/2}^3\left(1+\frac{1}{t^2}-\frac{2}{t}\right)dt$

$I=\frac{1}{16}\left(\frac{3}{2}+\frac{1}{3}+2\ln\frac{1}{2}\right)$

$I=\frac{1}{16}\left(\frac{11}{6}-2\ln 2\right)\Rightarrow 16I=\frac{11}{6}-\ln 4$

73. $\sin 3\theta=3\sin\theta-4\sin^3\theta=\sin\theta(1+2\cos 2\theta)\Rightarrow \sin x=\sin\frac{x}{3}\left(1+2\cos\frac{2x}{3}\right)$

$\sin\frac{x}{3}=\sin\left(\frac{x}{3^2}\right)\left(1+2\cos\frac{2x}{3^2}\right)$ and so on multiplying all

$$\sin x=\sin\left(\frac{x}{3^n}\right)\prod_{r=1}^{n}\left(1+2\cos\frac{2x}{3^r}\right)\Rightarrow\frac{\sin x}{x}=\lim_{n\to\infty}\frac{\sin\left(\frac{x}{3^n}\right)}{\left(\frac{x}{3^n}\right)}\cdot\prod_{r=1}^{n}\frac{\left(1+2\cos\frac{2x}{3^r}\right)}{3}\Rightarrow f(x)=\frac{\sin x}{x}$$

Thus $|xf(x)|+||x-2|-1|=|\sin x|+||x-2|-1|$ is not differentiable at 5 points in $(0,\,3\pi)$

74. Homogenize the equation

$\frac{x^2}{25}-\frac{y^2}{45}=\frac{(x\cos\theta+y\sin\theta)^2}{\lambda^2}$

Coefficient of x^2 + Coefficient $y^2=0$

$\Rightarrow\lambda=\pm 15/2$

$\Rightarrow$ equation of tangent to circle $C\equiv(0,0)$

as $x\cos\theta+y\sin\theta=\pm 15/2$

$r=15/2$

Equation of circle $x^2+y^2=\frac{225}{4}$

Length of tangent $(l)=\sqrt{25+50-\frac{225}{4}}=\frac{\sqrt{75}}{2}=4.33$

75. The two given families pass through A(2, 1) and B(19, –4) respectively

$\Rightarrow m_{OA}=\frac{1}{2}, m_{OB}=-\frac{4}{19}\Rightarrow$ Slope of $\perp$ lines are –2 and $\frac{19}{4}\Rightarrow\theta=\tan^{-1}\left(\frac{27}{34}\right)$

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